



अंतरिक्ष उपयोग केंद्र



SPACE APPLICATIONS CENTRE



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Subject	13 Scientist Engineer SC Statistics Mathematics Applied Mathematics

Section : Part A

Q.1 The particular integral of the following differential equation is

$$\frac{d^2y}{dx^2} + 16y = \sin 4x$$

- a. $-\frac{x}{4} \cos 4x$
- b. $-\frac{x}{8} \cos 4x$
- c. $\frac{x}{8} \sin 4x$
- d. $-\frac{x}{8} \cos 4x$

Ans

- ☒ A. a
- ☒ B. b
- ☒ C. c
- ☒ D. d

Question ID : 4776884512

Option 1 ID : 47768817125

Option 2 ID : 47768817126

Option 3 ID : 47768817127

Option 4 ID : 47768817128

Status : Answered

Chosen Option : D

Q.2 If $F(s)$ is the Fourier transform of $f(x)$ then which of the following is the Fourier transform of $f(ax)$, $a > 0$

- a. $\frac{1}{a} F(as)$
- b. $\frac{1}{a} F\left(\frac{s}{a}\right)$
- c. $aF\left(\frac{s}{a}\right)$
- d. $aF(as)$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884533
 Option 1 ID : 47768817209
 Option 2 ID : 47768817210
 Option 3 ID : 47768817211
 Option 4 ID : 47768817212
 Status : Answered
 Chosen Option : C

Q.3 Consider the space $C[0,1]$. Define for $f \in C[0,1]$

$$\|f\|_1 = \int_0^1 |f(x)| dx$$

$$\|f\|_\infty = \sup_{0 \leq x \leq 1} (|f(x)|)$$

Then which of the following statements is correct?

- a. $C[0,1]$ is complete with respect to $\|\cdot\|_\infty$ norm but not complete with respect to $\|\cdot\|_1$ norm
- b. $C[0,1]$ is complete with respect to both $\|\cdot\|_\infty$ norm and $\|\cdot\|_1$ norm.
- c. $C[0,1]$ is complete with respect to $\|\cdot\|_1$ norm but not complete with respect to $\|\cdot\|_\infty$ norm.
- d. $C[0,1]$ is not complete with respect to both $\|\cdot\|_\infty$ norm and $\|\cdot\|_1$ norm.

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884557
 Option 1 ID : 47768817305
 Option 2 ID : 47768817306
 Option 3 ID : 47768817307
 Option 4 ID : 47768817308
 Status : Answered
 Chosen Option : B

Q.4 Two unbiased coins are tossed. What is the probability of getting at most one head?

- a. $1/4$
- b. $1/2$
- c. $3/4$
- d. $1/3$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884515**
Option 1 ID : **47768817137**
Option 2 ID : **47768817138**
Option 3 ID : **47768817139**
Option 4 ID : **47768817140**
Status : **Answered**
Chosen Option : **C**

Q.5 There are 5 boxes, each containing 3 blue, 2 green and 1 red ball. One ball is drawn from each box randomly. Let X_1 denotes the random variable indicating the number of blue balls, and X_2 denotes the random variable indicating the number of red balls. Then, the probability that $X_1=2$ and $X_2=1$ is

- a. $7/36$
- b. $5/18$
- c. $1/4$
- d. $5/36$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884549**
Option 1 ID : **47768817273**
Option 2 ID : **47768817274**
Option 3 ID : **47768817275**
Option 4 ID : **47768817276**
Status : **Answered**
Chosen Option : **D**

Q.6 The radial acceleration of a particle moving in a plane curve is

- a. $\frac{1}{r} \frac{d}{dt} \left(r^2 \frac{d\theta}{dt} \right)$
- b. $\frac{d^2 r}{dt^2} - r \left(\frac{d\theta}{dt} \right)^2$
- c. $\frac{d}{dt} \left(r^2 \frac{d\theta}{dt} \right)$
- d. $\frac{d^2 r}{dt^2} - r^2 \frac{d\theta}{dt}$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884564

Option 1 ID : 47768817333

Option 2 ID : 47768817334

Option 3 ID : 47768817335

Option 4 ID : 47768817336

Status : Not Answered

Chosen Option : --

Q.7 $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a twice continuously differentiable scalar field such that $\text{div}(\nabla f) = 3$. Let S be the surface $x^2 + y^2 + z^2 = 1$ and $\hat{n}$ be the unit vector outward to S . Then the value of $\iint_S (\nabla f \cdot \hat{n}) ds$ is

- a. 4π
- b. 8π
- c. 2π
- d. 3π

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884538

Option 1 ID : 47768817229

Option 2 ID : 47768817230

Option 3 ID : 47768817231

Option 4 ID : 47768817232

Status : Answered

Chosen Option : A

Q.8 Number of finite abelian group of order 64 counted upto isomorphism are

- a. 6
- b. 8
- c. 10
- d. 11

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884521

Option 1 ID : 47768817161

Option 2 ID : 47768817162

Option 3 ID : 47768817163

Option 4 ID : 47768817164

Status : Answered

Chosen Option : A

Q.9 Let $f_n: [0,1] \rightarrow \mathbb{R}$, $n \geq 2$ be a sequence of continuous functions defined as follows:

$$f_n(x) = \begin{cases} 0, & \text{if } x \in [0, \frac{1}{2} - \frac{1}{n}] \\ \frac{n}{2}x + \left(\frac{2-n}{4}\right), & \text{if } \frac{1}{2} - \frac{1}{n} < x < \frac{1}{2} + \frac{1}{n} \\ 1, & \text{if } x \in [\frac{1}{2} + \frac{1}{n}, 1] \end{cases}$$

Define $\|f\|_1 = \int_0^1 |f(x)| dx$, $\|f\|_\infty = \sup_{0 \leq x \leq 1} (|f(x)|)$

Then which of the following statements is true

- a. $\{f_n\}$ is Cauchy in $\|\cdot\|_1$ norm but not Cauchy in $\|\cdot\|_\infty$ norm
- b. $\{f_n\}$ is Cauchy in both $\|\cdot\|_1$ and $\|\cdot\|_\infty$ norm
- c. $\{f_n\}$ is Cauchy in $\|\cdot\|_\infty$ norm but not Cauchy in $\|\cdot\|_1$ norm
- d. $\{f_n\}$ is not Cauchy in both $\|\cdot\|_1$ and $\|\cdot\|_\infty$ norm

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884565

Option 1 ID : 47768817337

Option 2 ID : 47768817338

Option 3 ID : 47768817339

Option 4 ID : 47768817340

Status : Not Answered

Chosen Option : --

Q.10 The solution to initial value problem $\frac{d^2x}{dt^2} - \frac{dx}{dt} = 1$, $x(0) =$

$0, \frac{dx}{dt}(0) = 1$ is

- a. $-3 - 2t + 3e^t$
- b. $-2 + 3t + 2e^{-t}$
- c. $-2 + t - 2e^t$
- d. $-1 - t + e^t$

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : **4776884562**

Option 1 ID : **47768817325**

Option 2 ID : **47768817326**

Option 3 ID : **47768817327**

Option 4 ID : **47768817328**

Status : **Answered**

Chosen Option : **B**

Q.11 If two events 'A' and 'B' are independent. Which of the following expressions is correct regarding their conditional probabilities?

- a. $P(A/B) = P(A \cap B)$
- b. $P(A/B) > P(A)$
- c. $P(A/B) < P(A)$
- d. $P(A/B) = P(A)$

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : **4776884552**

Option 1 ID : **47768817285**

Option 2 ID : **47768817286**

Option 3 ID : **47768817287**

Option 4 ID : **47768817288**

Status : **Answered**

Chosen Option : **D**

Q.12 Let X be a random variable with probability density function

$$f(x) = \begin{cases} \frac{1}{x^2}, & x \geq 1 \\ 0, & x < 1 \end{cases}$$

If $Y = \log_e X$, then $P(Y < 1 \mid Y < 2)$ equals

- a. $\frac{e}{1+e}$
- b. $\frac{e-1}{e+1}$
- c. $\frac{1}{1+e}$
- d. $\frac{1}{e-1}$

Ans  A. a

 B. b

 C. c

 D. d

Question ID : 4776884541

Option 1 ID : 47768817241

Option 2 ID : 47768817242

Option 3 ID : 47768817243

Option 4 ID : 47768817244

Status : Not Answered

Chosen Option : --

Q.13 The smallest degree of the polynomial that interpolates the data:

x	-2	-1	0	1	2
$f(x)$	3	1	1	3	7

- a. 1
- b. 3
- c. 4
- d. 2

Ans  A. a

 B. b

 C. c

 D. d

Question ID : 4776884532

Option 1 ID : 47768817205

Option 2 ID : 47768817206

Option 3 ID : 47768817207

Option 4 ID : 47768817208

Status : Answered

Chosen Option : D

Q.14 Consider the differential equation $x' = x^{1/3}$ with $x(0) = 0$. Then which of the following is correct about its solution?

- a. There exists unique solution
- b. There exist exactly two solutions
- c. There exist infinitely many solutions
- d. There exists no solution

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884573**
Option 1 ID : **47768817369**
Option 2 ID : **47768817370**
Option 3 ID : **47768817371**
Option 4 ID : **47768817372**
Status : **Answered**
Chosen Option : **A**

Q.15 Which of the following is correct decomposition into transpositions for the cycle (1,2,3) in S_3 (assuming left-to-right composition)?

- a. (2,1) (2,3)
- b. (1,3) (1,2)
- c. (2,3) (2,1)
- d. (1,2) (2,3)

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884501**
Option 1 ID : **47768817081**
Option 2 ID : **47768817082**
Option 3 ID : **47768817083**
Option 4 ID : **47768817084**
Status : **Answered**
Chosen Option : **B**

Q.16 $p(x) = \alpha + \beta(x+2) + \gamma(x+2)(x+1) + \delta(x+2)(x+1)(x)$
interpolates the following data

x	-2	-1	0	1	2
$f(x)$	2	-1	8	5	-34

Then $2(\alpha + \beta) + \gamma + \delta$

- a. 2
- b. 1
- c. 0
- d. (-)1

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884568**
 Option 1 ID : **47768817349**
 Option 2 ID : **47768817350**
 Option 3 ID : **47768817351**
 Option 4 ID : **47768817352**
 Status : **Answered**
 Chosen Option : **C**

Q.17 Let $f(x) = \begin{cases} 1, & |x| < 1 \\ 0, & |x| \geq 1 \end{cases}$ then the Fourier transform $F(s)$ of $f(x)$ is

- a. $\frac{2}{s} - \sin s$
- b. $\frac{\sin s}{s}$
- c. $\frac{2}{s} \sin s$
- d. $1 - \frac{\sin s}{s}$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884509**
 Option 1 ID : **47768817113**
 Option 2 ID : **47768817114**
 Option 3 ID : **47768817115**
 Option 4 ID : **47768817116**
 Status : **Not Answered**
 Chosen Option : --

- Q.18 $\lim_{z \rightarrow 0} \frac{z}{\bar{z}}$ is
- a. Does not exist
 - b. 1
 - c. -1
 - d. 0

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884534

Option 1 ID : 47768817213

Option 2 ID : 47768817214

Option 3 ID : 47768817215

Option 4 ID : 47768817216

Status : Answered

Chosen Option : B

- Q.19 Consider the following statements:
P: The set $\mathbb{N} \times \mathbb{N}$ is uncountable.
Q: The set $\{f: f \text{ is a function from } \mathbb{N} \text{ to } \{0, 1\}\}$ is uncountable.
- a. P is false but Q is true
 - b. Both P and Q are false
 - c. P is true but Q is false
 - d. Both P and Q are true

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884544

Option 1 ID : 47768817253

Option 2 ID : 47768817254

Option 3 ID : 47768817255

Option 4 ID : 47768817256

Status : Answered

Chosen Option : B

- Q.20** Consider the polynomial
 $f(x) = x^4 + 3x^3 + 6x^2 - 8x - 8, x \in \mathbb{R}$, then
- f has at least two real roots.
 - f has no real roots
 - f has exactly three complex roots.
 - f has at most one real root.

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : **4776884525**

Option 1 ID : **47768817177**

Option 2 ID : **47768817178**

Option 3 ID : **47768817179**

Option 4 ID : **47768817180**

Status : **Not Answered**

Chosen Option : --

- Q.21** Let $f(x) = \left\{ \frac{x}{2}, x \in (0, 2\pi) \right\}$ and has a period 2π . Let Fourier series representation of $f(x)$ is given by $\frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx]$. Then which of the following is true for $n \geq 1$

a. $a_n = \frac{1}{n}, b_n = (-1)^{\frac{1}{n}}$

b. $a_n = \frac{1}{n}, b_n = 0$

c. $a_n = 0, b_n = \frac{1}{n}$

d. $a_n = 0, b_n = (-1)^{\frac{1}{n}}$

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : **4776884546**

Option 1 ID : **47768817261**

Option 2 ID : **47768817262**

Option 3 ID : **47768817263**

Option 4 ID : **47768817264**

Status : **Not Answered**

Chosen Option : --

- Q.22** Let $f_n(x) = \frac{x^2}{x^2 + (1-nx)^2}$ $0 \leq x \leq 1$, $n = 1, 2, 3, \dots$ then which of the following statements is true?
- $\{f_n\}$ has a subsequence that can converge uniformly on $[0, 1]$
 - $\{f_n\}$ is uniformly bounded and equicontinuous
 - $\{f_n\}$ is uniformly bounded but not equicontinuous
 - $\{f_n\}$ is uniformly convergent on $[0, 1]$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884508
 Option 1 ID : 47768817109
 Option 2 ID : 47768817110
 Option 3 ID : 47768817111
 Option 4 ID : 47768817112
 Status : Answered
 Chosen Option : A

- Q.23** Let $\{e_n; n = 1, 2, 3, \dots\}$ be an orthonormal basis of a complex Hilbert Space H . Consider the following statements:
- P: There exists a bounded linear functional $f: H \rightarrow \mathbb{C}$ such that $f(e_n) = \frac{1}{n}$ for $n = 1, 2, 3, \dots$
- Q: There exists a bounded linear functional $g: H \rightarrow \mathbb{C}$ such that $g(e_n) = \frac{1}{n^2}$ for $n = 1, 2, 3, \dots$
- Both P and Q are false
 - Both P and Q are true
 - P is true and Q is false
 - P is false and Q is true

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884517
 Option 1 ID : 47768817145
 Option 2 ID : 47768817146
 Option 3 ID : 47768817147
 Option 4 ID : 47768817148
 Status : Answered
 Chosen Option : D

Q.24 Consider the fixed-point iteration

$$x_{n+1} = \phi(x_n) \quad n \geq 0 \quad \text{with} \quad \phi(x) = 3 + (x - 3)^2$$

Then the order of the convergence of the fixed-point iteration method is?

- a. 3
- b. 2
- c. 1
- d. 0

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884513

Option 1 ID : 47768817129

Option 2 ID : 47768817130

Option 3 ID : 47768817131

Option 4 ID : 47768817132

Status : Answered

Chosen Option : A

Q.25

$$A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}, \quad B = \begin{bmatrix} p^2 + q^2 & pr + qs \\ pr + qs & r^2 + s^2 \end{bmatrix}$$

where, p, q, r, s are non-zero real numbers if rank(A) = 1. What is the rank of B?

- a. 2
- b. 0
- c. 1
- d. None of the above

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884536

Option 1 ID : 47768817221

Option 2 ID : 47768817222

Option 3 ID : 47768817223

Option 4 ID : 47768817224

Status : Answered

Chosen Option : C

Q.26 A is a 4×4 real matrix such that $A^2 = A$. Then

- a. Rank of A can not be 4.
- b. Rank of A is either 1 or 2
- c. A is diagonalizable.
- d. A is zero matrix.

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884559

Option 1 ID : 47768817313

Option 2 ID : 47768817314

Option 3 ID : 47768817315

Option 4 ID : 47768817316

Status : Answered

Chosen Option : C

Q.27 If p is a real number less than 1, and two functions f and g belong to $L^p[0,1]$, then which of the following is true?

- a. $\|f + g\|_p \geq \|f\|_p + \|g\|_p$
- b. $\|f + g\|_p \leq \|f\|_p + \|g\|_p$
- c. $\|f + g\|_p < \|f\|_p + \|g\|_p$
- d. $\|f + g\|_p = \|f\|_p + \|g\|_p$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884507

Option 1 ID : 47768817105

Option 2 ID : 47768817106

Option 3 ID : 47768817107

Option 4 ID : 47768817108

Status : Answered

Chosen Option : B

Q.28 For the Poisson distribution, the mean is λ . What is the variance?

- a. λ
- b. λ^2
- c. $\sqrt{\lambda}$
- d. $\frac{1}{\lambda}$

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884554

Option 1 ID : 47768817293

Option 2 ID : 47768817294

Option 3 ID : 47768817295

Option 4 ID : 47768817296

Status : Answered

Chosen Option : A

Q.29 What values of x and y maximise the objective function $f(x,y) = 4x^2 + 3xy + 6y^2$ subject to constraint $x + y = 56$?

- a. $x=36, y=16$
- b. $x=20, y=36$
- c. $x=16, y=20$
- d. $x=36, y=20$

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884528

Option 1 ID : 47768817189

Option 2 ID : 47768817190

Option 3 ID : 47768817191

Option 4 ID : 47768817192

Status : Answered

Chosen Option : B

- Q.30** Harmonic conjugate $V(x, y)$ of $U(x, y) = 3x^2y + 2x^2 - y^3 - 2y$ is
- a. $3xy^2 + 4xy - x^3$
 - b. $3x^2y - 4xy + x^3$
 - c. $3xy^2 - 4xy + x^3$
 - d. $3xy^2 + 4x^2y - x^3$

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : **4776884555**

Option 1 ID : **47768817297**

Option 2 ID : **47768817298**

Option 3 ID : **47768817299**

Option 4 ID : **47768817300**

Status : **Not Answered**

Chosen Option : --

- Q.31** $F: \mathbb{R} \rightarrow \mathbb{R}$ is a twice continuously differentiable function with $F''(x) > 0$ for every $x \in \mathbb{R}$. Then, which of the following statements is incorrect.
- a. The function F can have at most one local minimum.
 - b. F can not have any local maxima
 - c. F' does not change sign over $\mathbb{R}$
 - d. F can not have inflection points.

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : **4776884574**

Option 1 ID : **47768817373**

Option 2 ID : **47768817374**

Option 3 ID : **47768817375**

Option 4 ID : **47768817376**

Status : **Not Answered**

Chosen Option : --

Q.32 A body falls from rest at a height 'h' above the surface of the earth under inverse square law $\frac{d^2x(t)}{dt^2} = -\frac{\mu}{(x(t))^2}$, $\mu > 0$, where $x(t)$ denotes the distance of the body from earth-centre at time t . The velocity of the body when it has just touched the earth is (assuming earth as a perfect sphere of radius R)

- a. $\sqrt{2\mu\left(\frac{1}{R} - \frac{1}{(R+h)}\right)}$
- b. $\sqrt{2\mu\frac{h}{R^2}}$
- c. $\sqrt{\frac{\mu h}{R^2}}$
- d. $\sqrt{\mu\left(\frac{1}{R} - \frac{1}{(R+h)}\right)}$

Ans ☒ A. a

☐ B. b

☐ C. c

☐ D. d

Question ID : 4776884577

Option 1 ID : 47768817385

Option 2 ID : 47768817386

Option 3 ID : 47768817387

Option 4 ID : 47768817388

Status : Answered

Chosen Option : A

Q.33 $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$ is

- a. $\frac{1}{e}$
- b. e
- c. e^2
- d. $\frac{1}{e^2}$

Ans ☐ A. a

☒ B. b

☐ C. c

☐ D. d

Question ID : 4776884506

Option 1 ID : 47768817101

Option 2 ID : 47768817102

Option 3 ID : 47768817103

Option 4 ID : 47768817104

Status : Answered

Chosen Option : B

Q.34 The solution of partial differential equation $u_x + u_t = 3u$; $u(0, t) = 4e^t$ is given by

- a. $u(x, t) = 4e^{2x+t}$
- b. $u(x, t) = 4e^{x+t}$
- c. $u(x, t) = 4e^{3x+t}$
- d. $u(x, t) = 4e^{-2x+t}$

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : **4776884539**

Option 1 ID : **47768817233**

Option 2 ID : **47768817234**

Option 3 ID : **47768817235**

Option 4 ID : **47768817236**

Status : **Answered**

Chosen Option : **C**

Q.35 For any group G of order 72, consider the following statements:

P: Number of 3-Sylow subgroups of G are either 1 or 4

Q: G has an element of order 2

- a. Both P and Q are false
- b. Both P and Q are true
- c. P is false but Q is true
- d. P is true but Q is false

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : **4776884540**

Option 1 ID : **47768817237**

Option 2 ID : **47768817238**

Option 3 ID : **47768817239**

Option 4 ID : **47768817240**

Status : **Not Answered**

Chosen Option : --

Q.36 Let $12^{11} = 11a + b$. Then, the value of b is

- a. 5
- b. 7
- c. 3
- d. 1

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884531

Option 1 ID : 47768817201

Option 2 ID : 47768817202

Option 3 ID : 47768817203

Option 4 ID : 47768817204

Status : Answered

Chosen Option : D

Q.37 Consider the sequence $s = \{s_k\}$ and $\{t_k\}$ such that

$$s_k = \sum_{n=1}^k \frac{1}{n^2}, \quad t_k = \sum_{n=1}^k \frac{1}{n} \quad \text{then}$$

- a. $\{s_k\}$ diverges but $\{t_k\}$ converges
- b. $\{s_k\}$ and $\{t_k\}$ both converges
- c. $\{s_k\}$ converges but $\{t_k\}$ diverges
- d. Both $\{s_k\}$ and $\{t_k\}$ diverges

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884567

Option 1 ID : 47768817345

Option 2 ID : 47768817346

Option 3 ID : 47768817347

Option 4 ID : 47768817348

Status : Answered

Chosen Option : C

Q.38 Consider the probability space (Ω, F, P) where $\Omega = [0, 2]$ and σ -Algebra $F = \{\emptyset, \Omega, [0, 1], (1, 2]\}$. Let X and Y be two functions on Ω defined as

$$X(\omega) = \begin{cases} 1 & \text{if } \omega \in [0, 1] \\ 2 & \text{if } \omega \in (1, 2] \end{cases}$$

and

$$Y(\omega) = \begin{cases} 2 & \text{if } \omega \in [0, 1.5] \\ 3 & \text{if } \omega \in (1.5, 2] \end{cases}$$

Then, which of the following is true?

- X is not a random variable with respect to F but Y is a random variable with respect to F
- X is a random variable with respect to F but Y is not a random variable with respect to F
- Neither of the X and Y is a random variable with respect to F
- Both X and Y are random variables with respect to F

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884560

Option 1 ID : 47768817317

Option 2 ID : 47768817318

Option 3 ID : 47768817319

Option 4 ID : 47768817320

Status : Not Answered

Chosen Option : --

Q.39 What is the principal value of i^i ?

a. $e^{-\frac{\pi}{2}}$

b. $e^{\frac{\pi}{2}}$

c. $e^{-\pi}$

d. e^{π}

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884548

Option 1 ID : 47768817269

Option 2 ID : 47768817270

Option 3 ID : 47768817271

Option 4 ID : 47768817272

Status : Answered

Chosen Option : D

- Q.40** Let $\vec{F} = (x + 2y)e^z\hat{i} + (ye^z + x^2)\hat{j} + y^2z\hat{k}$, and let S be the surface $x^2 + y^2 + z^2 = 1$, $z \geq 0$, if $\hat{n}$ is unit normal to S then $|\iint_S (\nabla \times \vec{F}) \cdot \hat{n} \, ds|$ is
- 3π
 - 2π
 - 4π
 - 8π

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884522

Option 1 ID : 47768817165

Option 2 ID : 47768817166

Option 3 ID : 47768817167

Option 4 ID : 47768817168

Status : Not Answered

Chosen Option : --

- Q.41** Let $y(x)$ be the solution of homogeneous differential equation $x^2y'' + 7xy' + 9y = 0$, $x > 0$ satisfying $y(1) = 1$. Then the value of $y(\frac{1}{2})$ is equal to
- 1
 - $\frac{1}{4}$
 - $\frac{1}{2}$
 - $\frac{1}{8}$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884556

Option 1 ID : 47768817301

Option 2 ID : 47768817302

Option 3 ID : 47768817303

Option 4 ID : 47768817304

Status : Not Answered

Chosen Option : --

- Q.42** If a hypothesis is not rejected at the 0.10 level of significance using a t-test, it
- must be rejected at the 0.05 level.
 - will not be rejected at the 0.05 level
 - may be rejected at the 0.05 level.
 - must be rejected at the 0.025 level.

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884558**
 Option 1 ID : **47768817309**
 Option 2 ID : **47768817310**
 Option 3 ID : **47768817311**
 Option 4 ID : **47768817312**
 Status : **Answered**
 Chosen Option : **C**

- Q.43** For what values of α and β , the quadrature formula $\int_{-1}^{+1} f(x) dx = \alpha f(-1) + \beta f(\beta)$ is exact for all polynomials of degree ≤ 1 (linear polynomials)?
- $\alpha = \frac{1}{2}, \beta = \frac{1}{2}$
 - $\alpha = 1, \beta = -1$
 - $\alpha = -1, \beta = -1$
 - $\alpha = 1, \beta = 1$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884569**
 Option 1 ID : **47768817353**
 Option 2 ID : **47768817354**
 Option 3 ID : **47768817355**
 Option 4 ID : **47768817356**
 Status : **Not Answered**
 Chosen Option : --

- Q.44** Suppose that
 $U = \mathbb{R}^2 \setminus \{(x, y) \in \mathbb{R}^2 : x, y \in \mathbb{N}\}$
 $V = \mathbb{R}^2 \setminus \{(x, y) \in \mathbb{R}^2 : x > 0, y = \frac{1}{x}\}$
 Then, with respect to Euclidean metric on $\mathbb{R}^2$, which of the following statements is true?
- U is disconnected but V is connected
 - Both U and V are connected
 - Both U and V are disconnected.
 - U is connected but V is disconnected

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884551**
 Option 1 ID : **47768817281**
 Option 2 ID : **47768817282**
 Option 3 ID : **47768817283**
 Option 4 ID : **47768817284**
 Status : **Answered**
 Chosen Option : **A**

- Q.45** Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -2 \\ 0 & -1 & 1 \end{bmatrix}$, let $A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$
 What is the trace of $(A - P^{-1}AP)$?
- 0
 - 2
 - 3
 - 5

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884530**
 Option 1 ID : **47768817197**
 Option 2 ID : **47768817198**
 Option 3 ID : **47768817199**
 Option 4 ID : **47768817200**
 Status : **Answered**
 Chosen Option : **D**

- Q.46** $\{f_n\}$ is a sequence of almost everywhere finite valued Lebesgue measurable functions on $[0,1]$ which converges to f almost uniformly. Consider the following two statements:
P: $\{f_n\}$ converges to f almost everywhere
Q: $\{f_n\}$ converges to f in measure
a. P is false but Q is true
b. P is true but Q is false
c. Both P and Q are true
d. Both P and Q are false

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884545
Option 1 ID : 47768817257
Option 2 ID : 47768817258
Option 3 ID : 47768817259
Option 4 ID : 47768817260
Status : Answered
Chosen Option : C

- Q.47** Which of the following is not a vector space
a. C over the field C
b. R over the field Q
c. R over the field C
d. C over the field R

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884550
Option 1 ID : 47768817277
Option 2 ID : 47768817278
Option 3 ID : 47768817279
Option 4 ID : 47768817280
Status : Answered
Chosen Option : C

Q.48 Let A be a $n \times n$ real matrix such that $AA^T = I_n$. Then which of the following is correct? ($\text{Det}(A)$ denotes the determinant of A)

- a. $\text{Det}(A) > 0$
- b. $\text{Det}(A) = \pm 1$
- c. $\text{Det}(A) = -1$
- d. $\text{Det}(A) = +1$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884500

Option 1 ID : 47768817077

Option 2 ID : 47768817078

Option 3 ID : 47768817079

Option 4 ID : 47768817080

Status : Answered

Chosen Option : D

Q.49 Consider the following statements:

P: Every sequence $\{x_n\}$ of real numbers has a monotone subsequence.

Q: Every bounded sequence $\{x_n\}$ of real numbers has a convergent subsequence.

Then,

- a. Both P and Q are false
- b. P is true but Q is false
- c. Both P and Q are true
- d. P is false but Q is true

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884504

Option 1 ID : 47768817093

Option 2 ID : 47768817094

Option 3 ID : 47768817095

Option 4 ID : 47768817096

Status : Answered

Chosen Option : C

Q.50 Newton's method

- a. can solve multivariable optimization problem
- b. can solve only single variable optimization problem
- c. can solve discrete optimization problem
- d. can solve stochastic optimization problem.

Ans ☒ A. a

☐ B. b

☐ C. c

☐ D. d

Question ID : 4776884511

Option 1 ID : 47768817121

Option 2 ID : 47768817122

Option 3 ID : 47768817123

Option 4 ID : 47768817124

Status : Not Answered

Chosen Option : --

Q.51 Consider the differential equation $\frac{dy}{dx} = x + y$, with initial condition $y(0)=1$. Using the Euler's method with a step size of $h=0.1$, the value of $y(0.2)$ is

- a. 1.2
- b. 1.22
- c. 1.32
- d. 1.12

Ans ☐ A. a

☒ B. b

☐ C. c

☐ D. d

Question ID : 4776884566

Option 1 ID : 47768817341

Option 2 ID : 47768817342

Option 3 ID : 47768817343

Option 4 ID : 47768817344

Status : Not Answered

Chosen Option : --

Q.52

Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. Then the eigen values of A are?

- a. 3, 1, -1
- b. 3, 1, 0
- c. 3, 0, 0
- d. 3, 0, -1

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884519

Option 1 ID : 47768817153

Option 2 ID : 47768817154

Option 3 ID : 47768817155

Option 4 ID : 47768817156

Status : Answered

Chosen Option : C

Q.53 Which of the following matrix is diagonalizable over R?

a. $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$

c. $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

b. $\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

d. $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884510

Option 1 ID : 47768817117

Option 2 ID : 47768817118

Option 3 ID : 47768817119

Option 4 ID : 47768817120

Status : Answered

Chosen Option : A

- Q.54** $T: (l^2, \|\cdot\|_2) \rightarrow (l^2, \|\cdot\|_2)$ be defined by $T(x_1, x_2, x_3, \dots) = (0, x_1, \frac{x_2}{2}, \frac{x_3}{3}, \dots)$. What is the norm of T ?
- $\|T\| = 0$
 - $\|T\| = \frac{1}{2}$
 - $\|T\| = 1$
 - $\|T\|$ is not defined

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : **4776884529**

Option 1 ID : **47768817193**

Option 2 ID : **47768817194**

Option 3 ID : **47768817195**

Option 4 ID : **47768817196**

Status : **Answered**

Chosen Option : **A**

- Q.55** Gauss-Seidel method is used to solve the following system of linear equations:
- $$2x_1 + x_2 = 4$$
- $$2x_2 + x_3 = 7$$
- $$2x_2 + 3x_3 = 13$$
- with initial condition $(x_1, x_2, x_3) = (1, 1, 1)$. What is the absolute error in approximating x_3 at the first iterate?

a. $7/3$

b. $2/3$

c. $1/3$

d. $5/3$

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : **4776884575**

Option 1 ID : **47768817377**

Option 2 ID : **47768817378**

Option 3 ID : **47768817379**

Option 4 ID : **47768817380**

Status : **Not Answered**

Chosen Option : **--**

Q.56 Area bounded by the curves $y = 3 - x^2$ and $y = 2x$ is?

- a. 16
- b. $\frac{16}{3}$
- c. $\frac{64}{3}$
- d. $\frac{32}{3}$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884514**

Option 1 ID : **47768817133**

Option 2 ID : **47768817134**

Option 3 ID : **47768817135**

Option 4 ID : **47768817136**

Status : **Not Answered**

Chosen Option : --

Q.57 $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = x^6 - 2x^2y - x^4y + 2y^2$, then which of the following statements is true?

- a. The origin is not a critical point of f
- b. f has a saddle point at origin
- c. f has a local minimum at origin
- d. f has a local maximum at origin

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884503**

Option 1 ID : **47768817089**

Option 2 ID : **47768817090**

Option 3 ID : **47768817091**

Option 4 ID : **47768817092**

Status : **Answered**

Chosen Option : **B**

Q.58 Consider the function f defined on interval $[-1,1]$ as follows

$$f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

Then

- a. $f''(0)$ exists
- b. $f'(0)$ exists and f' is bounded
- c. f and f' are continuous
- d. $f'(0)$ exists but f' is not bounded

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884571

Option 1 ID : 47768817361

Option 2 ID : 47768817362

Option 3 ID : 47768817363

Option 4 ID : 47768817364

Status : Answered

Chosen Option : D

Q.59 The stable critical points of the differential equation $x' = \sin(x)$ are

- a. $x = k\pi$, k is an odd integer
- b. $x = k\pi$, k is an integer
- c. $x = k\pi$, k is even integer
- d. $x = k\pi$, $k = 0$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884542

Option 1 ID : 47768817245

Option 2 ID : 47768817246

Option 3 ID : 47768817247

Option 4 ID : 47768817248

Status : Answered

Chosen Option : C

Q.60 A dice is thrown two times and the sum of scores appearing on the dice is observed to be a multiple of 4. Then the conditional probability that the score 4 has appeared exactly twice is

- a. $\frac{1}{3}$
- b. $\frac{1}{9}$
- c. $\frac{1}{5}$
- d. $\frac{1}{8}$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884526

Option 1 ID : 47768817181

Option 2 ID : 47768817182

Option 3 ID : 47768817183

Option 4 ID : 47768817184

Status : Answered

Chosen Option : B

Q.61 The wait-time at a restaurant follows an exponential distribution with probability density function $f(x) = \frac{1}{3} e^{-\frac{x}{3}}$. What is the probability that the wait-time takes more than 5 minutes?

- a. $e^{-\frac{5}{3}}$
- b. $e^{-\frac{1}{3}}$
- c. e^{-15}
- d. $\frac{1}{3} e^{-5}$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884518

Option 1 ID : 47768817149

Option 2 ID : 47768817150

Option 3 ID : 47768817151

Option 4 ID : 47768817152

Status : Not Answered

Chosen Option : --

Q.62 Let $\phi(x)$ denotes the cumulative distribution function of a standard normal random variable. If a random variable X has the cumulative distribution function

$$F(x) = \begin{cases} \phi(x), & x < -2 \\ \phi(x+2), & x \geq -2 \end{cases}$$

Then which of the following statements is true?

- a. $P(x = -2) = \frac{1}{2}$
- b. $P(x \leq -2) = \frac{1}{2}$
- c. $P(x < -2) = \frac{1}{2}$
- d. $P(x \leq 0) = \frac{1}{2}$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884561

Option 1 ID : 47768817321

Option 2 ID : 47768817322

Option 3 ID : 47768817323

Option 4 ID : 47768817324

Status : Not Answered

Chosen Option : --

Q.63 The value of $(1 + \sqrt{3}i)^{2011}$ is

- a. $2^{2011}(1 + \sqrt{3}i)$
- b. $2^{2010}(1 + \sqrt{3}i)$
- c. $2^{2012}(1 + \sqrt{3}i)$
- d. $2^{2010}(1 - \sqrt{3}i)$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884520

Option 1 ID : 47768817157

Option 2 ID : 47768817158

Option 3 ID : 47768817159

Option 4 ID : 47768817160

Status : Not Answered

Chosen Option : --

Q.64 If $P(A) = \frac{2}{3}$, $P(B) = \frac{5}{8}$ and the probability of only A occurring is $\frac{1}{6}$.
What is the probability of only B occurring?

- a. $\frac{1}{2}$
- b. $\frac{1}{9}$
- c. $\frac{1}{8}$
- d. $\frac{1}{6}$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884535

Option 1 ID : 47768817217

Option 2 ID : 47768817218

Option 3 ID : 47768817219

Option 4 ID : 47768817220

Status : Not Answered

Chosen Option : --

Q.65 Which of the following statements is not true?
a. The determinant of an orthonormal matrix is either +1 or -1.
b. The determinant of an orthonormal matrix can be negative.
c. Every orthonormal matrix is a rotation matrix.
d. Every rotation matrix is an orthonormal matrix.

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884523

Option 1 ID : 47768817169

Option 2 ID : 47768817170

Option 3 ID : 47768817171

Option 4 ID : 47768817172

Status : Answered

Chosen Option : A

Q.66 The system of equations

$$X + Y + Z = 5$$

$$2X + 4Y + 6Z = 20$$

$$X + 2Y + \lambda Z = \mu$$

has infinitely many solutions for what value of λ and μ ?

- a. $\lambda \neq 3, \mu \neq 10$
- b. $\lambda \neq 3, \mu = 10$
- c. $\lambda = 3, \mu \neq 10$
- d. $\lambda = 3, \mu = 10$

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884537

Option 1 ID : 47768817225

Option 2 ID : 47768817226

Option 3 ID : 47768817227

Option 4 ID : 47768817228

Status : Not Answered

Chosen Option : --

Q.67

The type of partial differential equation $\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 p}{\partial y^2} + 3\frac{\partial^2 p}{\partial x \partial y} + 2\frac{\partial p}{\partial x} -$

$$\frac{\partial p}{\partial y} = 0$$

- a. Parabolic
- b. Hyperbolic
- c. Elliptic
- d. None of the above

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884578

Option 1 ID : 47768817389

Option 2 ID : 47768817390

Option 3 ID : 47768817391

Option 4 ID : 47768817392

Status : Answered

Chosen Option : A

Q.68 If the trapezoidal rule with single interval $[0,1]$ is exact for approximating the integral $\int_0^1 (x^3 - cx^2)dx$. Then the value of $2c$ is equal to

- a. 5
- b. 1
- c. 2
- d. 3

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884543**
 Option 1 ID : **47768817249**
 Option 2 ID : **47768817250**
 Option 3 ID : **47768817251**
 Option 4 ID : **47768817252**
 Status : **Not Answered**
 Chosen Option : --

Q.69 Let $h(x)$ be twice continuously differential function in interval $[1,2]$ with fixed point α , then the sequence of iterates $x_{n+1} = h(x_n)$ converges to α quadratically, provided

- a. $\frac{dh}{dx}(\alpha) \neq 0$
- b. $\frac{dh}{dx}(\alpha) \neq 0, \frac{d^2h}{dx^2}(\alpha) = 0$
- c. $\frac{dh}{dx}(\alpha) \neq 0, \frac{d^2h}{dx^2}(\alpha) \neq 0$
- d. $\frac{dh}{dx}(\alpha) = 0, \frac{d^2h}{dx^2}(\alpha) \neq 0$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884563**
 Option 1 ID : **47768817329**
 Option 2 ID : **47768817330**
 Option 3 ID : **47768817331**
 Option 4 ID : **47768817332**
 Status : **Not Answered**
 Chosen Option : --

Q.70 Let $T: \mathbb{R}^4 \rightarrow \mathbb{R}^2$ be a linear operator such that $\text{rank}(T) = 1$. What is the dimension of null space of T ?

- a. 1
- b. 3
- c. 4
- d. 2

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884505

Option 1 ID : 47768817097

Option 2 ID : 47768817098

Option 3 ID : 47768817099

Option 4 ID : 47768817100

Status : Answered

Chosen Option : B

Q.71 What is the value of $\int_C \frac{1}{z^2+4} dz$, where C is the circle $|z - i| = 2$ oriented counter-clockwise

- a. $\frac{\pi}{2}$
- b. $(-)\frac{\pi}{2}$
- c. $\frac{\pi}{4}$
- d. $i\frac{\pi}{2}$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884570

Option 1 ID : 47768817357

Option 2 ID : 47768817358

Option 3 ID : 47768817359

Option 4 ID : 47768817360

Status : Not Answered

Chosen Option : --

Q.72 A solution for the differential equation $\frac{dy}{dx} = |x|$, $x \in \mathbb{R}$ is given by

- a. $y = \frac{x^2}{2} + x$
- b. $y = \frac{x^2}{2}$
- c. $y = \frac{x|x|}{2}$
- d. $y = (-)\frac{x|x|}{2}$

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884579**

Option 1 ID : **47768817393**

Option 2 ID : **47768817394**

Option 3 ID : **47768817395**

Option 4 ID : **47768817396**

Status : **Not Answered**

Chosen Option : --

Q.73 Consider the following statements for $x \in \mathbb{R}$ and $y \in \mathbb{R}$

P: $d_1(x, y) = (x-y)^2$ is a metric on $\mathbb{R}$

Q: $d_2(x, y) = \sqrt{|x - y|}$ is a metric on $\mathbb{R}$

Then,

- a. Both Q and P are False
- b. Q is True but P is False.
- c. Both Q and P are True.
- d. Q is False but P is True.

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884576**

Option 1 ID : **47768817381**

Option 2 ID : **47768817382**

Option 3 ID : **47768817383**

Option 4 ID : **47768817384**

Status : **Not Answered**

Chosen Option : --

Q.74 The radial velocity of a particle moving on a circular path of radius a is

- a. $a \frac{d^2\theta}{dt^2}$
- b. $\frac{d\theta}{dt}$
- c. $a \frac{d\theta}{dt}$
- d. 0

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884547
Option 1 ID : 47768817265
Option 2 ID : 47768817266
Option 3 ID : 47768817267
Option 4 ID : 47768817268
Status : Answered
Chosen Option : C

Q.75 If the characteristic polynomial f_A for a matrix A is given by $f_A(x) = (x-1)^2(x-2)$ then the number of possible Jordan forms of A is

- a. 3
- b. 2
- c. 1
- d. 4

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884516
Option 1 ID : 47768817141
Option 2 ID : 47768817142
Option 3 ID : 47768817143
Option 4 ID : 47768817144
Status : Answered
Chosen Option : A

- Q.76** Which of the following statements is correct?
- Every cumulative distribution function F is increasing and right-continuous.
 - Every cumulative distribution function F is non-decreasing and left-continuous.
 - Every cumulative distribution function F is non-decreasing and right-continuous.
 - Every cumulative distribution function F is increasing and left-continuous.

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884527**
 Option 1 ID : **47768817185**
 Option 2 ID : **47768817186**
 Option 3 ID : **47768817187**
 Option 4 ID : **47768817188**
 Status : **Not Answered**
 Chosen Option : --

- Q.77** Let Q be the set of all rational numbers with metric $(p, q) = |p - q|$. Let $E = \{p \in Q \mid 2 < p^2 < 3\}$, then which of the following statements is true ?
- E is compact in Q
 - E is closed and bounded in Q but E is not compact in Q
 - E is not closed but bounded in Q
 - E is neither closed nor bounded in Q

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : **4776884524**
 Option 1 ID : **47768817173**
 Option 2 ID : **47768817174**
 Option 3 ID : **47768817175**
 Option 4 ID : **47768817176**
 Status : **Not Answered**
 Chosen Option : --

Q.78 The system of linear equations

$$\begin{bmatrix} 2 & 1 & 3 \\ 4 & 0 & 1 \\ 0 & 2 & 5 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 5 \\ (-)4 \\ 14 \end{bmatrix} \text{ has}$$

- a. Unique solution
- b. Infinitely many solutions
- c. No solution
- d. Exactly two solutions

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884553

Option 1 ID : 47768817289

Option 2 ID : 47768817290

Option 3 ID : 47768817291

Option 4 ID : 47768817292

Status : Not Answered

Chosen Option : --

Q.79 Solution of the differential equation $\frac{dy}{dx} + \frac{2x}{y} = 0$ represents a family of

- a. Hyperbolas
- b. Circles
- c. Parabolas
- d. Ellipses

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884572

Option 1 ID : 47768817365

Option 2 ID : 47768817366

Option 3 ID : 47768817367

Option 4 ID : 47768817368

Status : Not Answered

Chosen Option : --

Q.80 Let G be an abelian group of order 24, then the number of subgroups of order 8 in G are

- a. 3
- b. 2
- c. 1
- d. 0

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884502
Option 1 ID : 47768817085
Option 2 ID : 47768817086
Option 3 ID : 47768817087
Option 4 ID : 47768817088
Status : Not Answered
Chosen Option : --

Section : Part I

Q.1 All Engineers are logical thinkers. Some logical thinkers are musicians. Which of the following statement is correct:

- a. All engineers are musicians
- b. Some engineers can be musician
- c. No engineer is musician
- d. All musicians are engineers.

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884589
Option 1 ID : 47768817433
Option 2 ID : 47768817434
Option 3 ID : 47768817435
Option 4 ID : 47768817436
Status : Answered
Chosen Option : B

Q.2 Find the next in sequence

AB, DE, GH, JK, ____

- a. LM
- b. MN
- c. OP
- d. QR

Ans  A. a
 B. b
 C. c
 D. d

Question ID : 4776884588
Option 1 ID : 47768817429
Option 2 ID : 47768817430
Option 3 ID : 47768817431
Option 4 ID : 47768817432
Status : Answered
Chosen Option : B

Q.3 Find the next number in sequence.

2, 6, 7, 21, 22, ____

- a. 44
- b. 46
- c. 66
- d. 68

Ans  A. a
 B. b
 C. c
 D. d

Question ID : 4776884581
Option 1 ID : 47768817401
Option 2 ID : 47768817402
Option 3 ID : 47768817403
Option 4 ID : 47768817404
Status : Answered
Chosen Option : C

Q.4 All cats are animals. Some animals are black. What can be concluded?

- a. All cats are black
- b. Some cats may be black
- c. No cats are black
- d. All animals are cats

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884583
Option 1 ID : 47768817409
Option 2 ID : 47768817410
Option 3 ID : 47768817411
Option 4 ID : 47768817412
Status : Answered
Chosen Option : B

Q.5 Find the next number in sequence.

3, 9, 27, 81, ____

- a. 162
- b. 196
- c. 243
- d. 324

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884580
Option 1 ID : 47768817397
Option 2 ID : 47768817398
Option 3 ID : 47768817399
Option 4 ID : 47768817400
Status : Answered
Chosen Option : C

Q.6 Book : Reading :: Knife : ____

- a. Writing
- b. Cutting
- c. Eating
- d. Drawing

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884587

Option 1 ID : 47768817425

Option 2 ID : 47768817426

Option 3 ID : 47768817427

Option 4 ID : 47768817428

Status : Answered

Chosen Option : B

Q.7 If yesterday was Monday, what day will it be 3 days after tomorrow?

- a. Friday
- b. Saturday
- c. Sunday
- d. Monday

Ans ☒ A. a

☒ B. b

☒ C. c

☒ D. d

Question ID : 4776884585

Option 1 ID : 47768817417

Option 2 ID : 47768817418

Option 3 ID : 47768817419

Option 4 ID : 47768817420

Status : Answered

Chosen Option : B

Q.8 Find the odd one out

- a. Apple
- b. Banana
- c. Carrot
- d. Mango

Ans  A. a
 B. b
 C. c
 D. d

Question ID : 4776884584
Option 1 ID : 47768817413
Option 2 ID : 47768817414
Option 3 ID : 47768817415
Option 4 ID : 47768817416
Status : Answered
Chosen Option : C

Q.9 Find the next in sequence.

Circle → Square → Triangle → Circle → Square → ____

- a. Circle
- b. Triangle
- c. Rectangle
- d. Pentagon

Ans  A. a
 B. b
 C. c
 D. d

Question ID : 4776884586
Option 1 ID : 47768817421
Option 2 ID : 47768817422
Option 3 ID : 47768817423
Option 4 ID : 47768817424
Status : Answered
Chosen Option : D

Q.10 The ratio of boys to girls is 3:2 in a class of 30 students. How many girls are there?

- a. 10
- b. 12
- c. 15
- d. 18

Ans  A. a
 B. b
 C. c
 D. d

Question ID : **4776884582**
Option 1 ID : **47768817405**
Option 2 ID : **47768817406**
Option 3 ID : **47768817407**
Option 4 ID : **47768817408**
Status : **Answered**
Chosen Option : **B**

Section : **Part II**

Q.11 DOG = 26, CAT = 24. What is BAT?

- a. 21
- b. 22
- c. 23
- d. 24

Ans  A. a
 B. b
 C. c
 D. d

Question ID : **4776884592**
Option 1 ID : **47768817445**
Option 2 ID : **47768817446**
Option 3 ID : **47768817447**
Option 4 ID : **47768817448**
Status : **Answered**
Chosen Option : **C**

Q.12 If $A \rightarrow B$ means A is mother of B and $A - B$ means A is brother of B, what does $P \rightarrow Q - R$ mean?

- a. P is sister of R
- b. P is mother of Q and Q is brother of R
- c. Q is mother of P
- d. R is mother of Q

Ans  A. a
 B. b
 C. c
 D. d

Question ID : 4776884591
Option 1 ID : 47768817441
Option 2 ID : 47768817442
Option 3 ID : 47768817443
Option 4 ID : 47768817444
Status : Answered
Chosen Option : B

Q.13 All roses are flowers. All flowers need water. What follows?

- a. Roses don't need water
- b. All roses need water
- c. Only roses need water
- d. Some flowers don't need water

Ans  A. a
 B. b
 C. c
 D. d

Question ID : 4776884593
Option 1 ID : 47768817449
Option 2 ID : 47768817450
Option 3 ID : 47768817451
Option 4 ID : 47768817452
Status : Answered
Chosen Option : B

Q.14 A product priced at ₹500 is increased by 20% and then discounted by 20%. What is the final price?

- a. ₹400
- b. ₹480
- c. ₹500
- d. ₹520

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884594

Option 1 ID : 47768817453

Option 2 ID : 47768817454

Option 3 ID : 47768817455

Option 4 ID : 47768817456

Status : Answered

Chosen Option : C

Q.15 A, B, C, D, E are in a row. A is to the left of B and to the right of D. B is between A and C. E is on the left of D. Who is in middle?

- a. A
- b. B
- c. C
- d. D

Ans ☒ A. a
☒ B. b
☒ C. c
☒ D. d

Question ID : 4776884590

Option 1 ID : 47768817437

Option 2 ID : 47768817438

Option 3 ID : 47768817439

Option 4 ID : 47768817440

Status : Answered

Chosen Option : A